- `Rb ≤ Ny ≤ Rt`: we need to traverse both branches;
- `Ny > Rt`: we can prune the left branch;
- `Rb > Ny`: we can prune the right branch.

And similarly for vertical splits, checking the x coordinates: likewise, it can be generalized for k-dimensional spaces, cycling through dimensions.

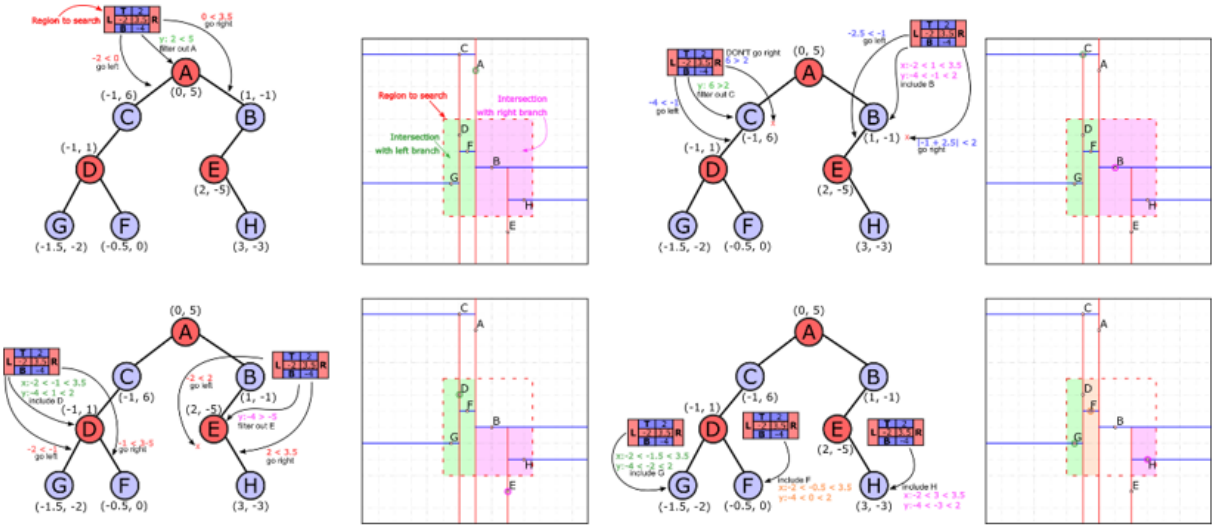


Figure 9.26 Region Search on a k-d tree: returns all the points in a k-d tree within a given hyper-rectangle. This means looking for points that, for each coordinate, satisfy two inequalities: each coordinate must be within a range; for instance, in this 2-D example, points' x coordinates need to be between `-2` and `3.5`, and y coordinates between `-4` and `2`.

This method is particularly useful when we need to search values within simple boundaries for each feature in data. For instance, if we have a dataset containing tenure and salary of employees, we might want to search all employees who